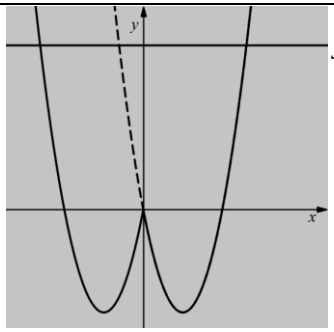
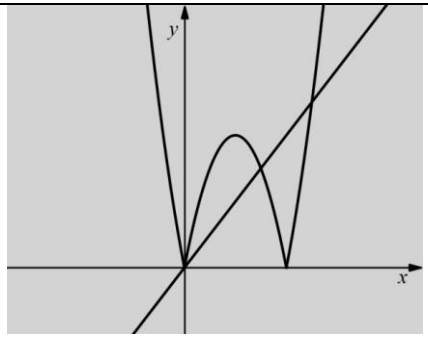
1	$3\tan^2\theta + 7\sec\theta - 3 = 0 \Rightarrow 3\left(\sec^2\theta - 1\right) + 7\sec\theta - 3 = 0$	M1
	$3\sec^2\theta + 7\sec\theta - 6 = 0$	A1
	$\left(3\sec\theta - 2\right)\left(\sec\theta + 3\right) = 0 \Rightarrow \sec\theta = ... \Rightarrow \cos\theta = ...$	dM1
	$\theta = 109.5°, 250.5°$	A1, A1
		(5)
		Total 5

2 (a)	$x = 2y^2 + 5y - 6$	
	$\frac{dx}{dy} = 4y + 5 \Rightarrow \frac{dy}{dx} = \frac{1}{4y+5}$	M1, A1
		(2)
(b)	Sets their $4y + 5 = 0 \Rightarrow y = -\frac{5}{4}$	M1
	Substitutes their $y = -\frac{5}{4}$ into $x = 2y^2 + 5y - 6$ to find x	dM1
	$\left(-\frac{73}{8}, -\frac{5}{4}\right)$	A1
		(3)
		Total 5

3 (a)	 Graphical interpretation of $\mathrm{f}(\lvert x \rvert) = 48$	
	One of $8, -8$	B1
	Attempts to solve an appropriate equation E.g. $2x^2 - 10x = 48 \Rightarrow x^2 - 5x - 24 = 0 \Rightarrow (x \pm 8)(x \pm 3) = 0 \Rightarrow x = ...$	M1
	$x = 8, -8$ with no additional values	A1
		(3)
(b)	 $y = \frac{5}{2}x$ Graphical interpretation of $\lvert \mathrm{f}(x) \rvert \ldots \frac{5}{2}x$	
	Attempts to solve $2x^2 - 10x = \frac{5}{2}x \Rightarrow 4x^2 - 25x = 0 \Rightarrow x = \frac{25}{4}$ OR attempts to solve (o.e.) $10x - 2x^2 = \frac{5}{2}x \Rightarrow 4x^2 - 15x = 0 \Rightarrow x = \frac{15}{4}$	M1
	Attempts to solve $2x^2 - 10x = \frac{5}{2}x \Rightarrow 4x^2 - 25x = 0 \Rightarrow x = \frac{25}{4}$ AND attempts to solve (o.e.) $10x - 2x^2 = \frac{5}{2}x \Rightarrow 4x^2 - 15x = 0 \Rightarrow x = \frac{15}{4}$	dM1
	Achieves both critical values $x = \frac{15}{4}$, $x = \frac{25}{4}$	A1
	Correct set of values x,, $\frac{15}{4}$ or $x \ldots \frac{25}{4}$	A1
		(4)
		Total 7

4 (a)	$\log_{10} N = 2 \Rightarrow N = 100$	B1
		(1)
(b)	$\log_{10} N = 0.35t + 2 \Rightarrow N = 10^{0.35t+2}$	M1
	$\Rightarrow N = 10^{0.35t} \times 10^{2}$	dM1
	$\Rightarrow N = 100 \times 2.24^{t}$	A1
		(3)
(c)	Rate of growth at 5 hours is $\frac{dN}{dt} = 100 \times \ln 2.24 \times 2.24^{5}$	M1
	Allow an answer in the range 4530 to 4550 (bacteria per hour)	A1
		(2)
		Total 6

5(a)	$\sin 3x = \sin(2x+x) = \sin 2x\cos x + \cos 2x \sin x$	M1
	$= (2\sin x\cos x)\cos x + (1-2\sin^2 x)\sin x$	dM1
	$= 2\sin x\cos^2 x + \sin x - 2\sin^3 x$	A1
	$= 2\sin x(1-\sin^2 x) + \sin x - 2\sin^3 x = 3\sin x - 4\sin^3 x$	A1
		(4)
(b)	$2\sin 3\theta = 5\sin 2\theta \Rightarrow 2(3\sin\theta - 4\sin^3\theta) = 10\sin\theta\cos\theta$	M1
	Divides or takes out $\sin\theta$ as a factor and uses $\sin^2\theta = 1-\cos^2\theta$ to set up and solve a 3TQ in $\cos\theta$ E.g. $\Rightarrow 6\sin\theta - 8\sin^3\theta = 10\sin\theta\cos\theta \Rightarrow 6-8(1-\cos^2\theta) = 10\cos\theta$ $\Rightarrow 4\cos^2\theta - 5\cos\theta - 1 = 0$ $\Rightarrow \cos\theta = \dfrac{5-\sqrt{41}}{8} = (-0.175...)$	dM1
	Any two of the following four answers $\sin\theta = 0 \Rightarrow \theta = 180°, 360°$ $\cos\theta = \dfrac{5-\sqrt{41}}{8} \Rightarrow \theta =$ awrt 100° or awrt 260°	A1
	All of 180°, 360°, awrt 100.1°, awrt 259.9°	A1
		(4)
		Total 8

6 (a)	$\text{gf}(2) = \text{g}(3) = 3^2 + 5$	M1
	$= 14$	A1
		(2)
(b)	Correct attempt at inverse $y = 6 - \frac{21}{2x+3} \Rightarrow 2x+3 = \frac{21}{6-y} \Rightarrow x = \left(\frac{21}{6-y} - 3\right) \div 2$	M1
	$\text{f}^{-1}(x) = \frac{3x+3}{2(6-x)}$ or $\text{f}^{-1}(x) = \frac{21}{2(6-x)} - \frac{3}{2}$	A1
	-1 ,, $x < 6$	B1
		(3)
(c)	$\text{gg}(x) = 126 \Rightarrow \left(x^2+5\right)^2 + 5 = 126 \Rightarrow x^2 + 5 = \sqrt{121}$	M1
	$\Rightarrow x^2 = "6" \Rightarrow x = (\pm)\sqrt{6}$	dM1
	$\Rightarrow x = \pm\sqrt{6}$	A1
		(3)
		Total 8

7 (a)	$f(x)=x^3\sqrt{4x+7} \Rightarrow f'(x)=3x^2\sqrt{4x+7}+2x^3\left(4x+7\right)^{-\frac{1}{2}}$	M1, A1
	$\Rightarrow f'(x)=3x^2\sqrt{4x+7}+\dfrac{2x^3}{\sqrt{4x+7}}=\dfrac{3x^2\left(4x+7\right)+2x^3}{\sqrt{4x+7}}$	dM1
	$\Rightarrow f'(x)=\dfrac{7x^2\left(2x+3\right)}{\sqrt{4x+7}}$	A1
		(4)
(b)	Substitutes $x=-\dfrac{3}{2}$ into $x^3\sqrt{4x+7} \Rightarrow y=...$	M1
	$\left(-\dfrac{3}{2},-\dfrac{27}{8}\right)$	A1
		(2)
(c)	Attempts $-4\times"-\dfrac{27}{8}"$	M1
	y ,, $\dfrac{27}{2}$	A1
		(2)
(d)	$\left(2,7\right)$	M1, A1
		(2)
		Total 10

8 (a)	52 b.p.m.	B1
		(1)
(b)	32 b.p.m.	B1
		(1)
(c)	$\dfrac{\mathrm{d}H}{\mathrm{d}t} = -8\,\mathrm{e}^{-0.2t} + 18\,\mathrm{e}^{-0.9t}$	M1, A1
	Sets $-8\,\mathrm{e}^{-0.2t} + 18\,\mathrm{e}^{-0.9t} = 0 \Rightarrow 4\mathrm{e}^{0.7t} = 9$	dM1
	$\Rightarrow 0.7t = \ln\dfrac{9}{4} \Rightarrow t = ...$	M1
	T = 1.158 (minutes)	A1
		(5)
(d)	$37 = 32 + 40\,\mathrm{e}^{-0.2t} - 20\,\mathrm{e}^{-0.9t} \Rightarrow \mathrm{e}^{-0.2t} = \dfrac{1+4\,\mathrm{e}^{-0.9t}}{8}$	M1
	$\Rightarrow \mathrm{e}^{0.2t} = \dfrac{8}{1+4\,\mathrm{e}^{-0.9t}} \Rightarrow t = 5\ln\left(\dfrac{8}{1+4\,\mathrm{e}^{-0.9t}}\right)$	A1*
		(2)
(e)	$t_2 = 5\ln\left(\dfrac{8}{1+4\mathrm{e}^{-0.9\times10}}\right) = ...$	M1
	$\left(t_2\right)$ = awrt 10.3947	A1
	(M)= 10.3955	A 1
		(3)
		Total 12

9 (a)	$f'(x) = \dfrac{(2x+1)\times(12x+4)-2\left(6x^2+4x-2\right)}{(2x+1)^2}$	M1 A1
	$= \dfrac{12x^2+12x+8}{(2x+1)^2}$ o.e	A1
		(3)
(b)	At $x=2 \Rightarrow f'(x) = \dfrac{12\times 2^2+12\times 2+8}{(2\times 2+1)^2} = \left(\dfrac{16}{5}\right)$	M1
	Full method of normal $y-6=-\dfrac{5}{16}(x-2)$	dM1
	$16y-96=-5x+10 \Rightarrow \quad 16y+5x=106 \quad *$	A1*
		(3)
(c)	Any correct value $A=3,\ B=\dfrac{1}{2}$ *or* $D=-\dfrac{5}{2}$	B1
	$6x^2+4x-2=(Ax+B)(2x+1)+D \Rightarrow$ Values of A, B and D	M1
	$3x+\dfrac{1}{2}+\dfrac{-5/2}{2x+1}$ o.e	A1
		(3)
(d)	$\displaystyle\int 3x+\frac{1}{2}+\frac{-5/2}{2x+1}\,dx = \frac{3}{2}x^2+\frac{1}{2}x-\frac{5}{4}\ln(2x+1)$	M1, A1 ft
	Area under curve = $\left[\dfrac{3}{2}x^2+\dfrac{1}{2}x-\dfrac{5}{4}\ln(2x+1)\right]_{\frac{1}{3}}^{2} = \left(\dfrac{3}{2}\times 2^2+\dfrac{1}{2}\times 2-\dfrac{5}{4}\ln(5)\right)-\left(\dfrac{3}{2}\times\left(\dfrac{1}{3}\right)^2+\dfrac{1}{2}\times\left(\dfrac{1}{3}\right)-\dfrac{5}{4}\ln\left(\dfrac{5}{3}\right)\right)$	dM1
	Area of R = $\displaystyle\int_{\frac{1}{3}}^{2} 3x+\frac{1}{2}+\frac{-5/2}{2x+1}\,dx+\frac{1}{2}\times 6\times\left(\frac{106}{5}-2\right)$	M1
	Area of R = $\dfrac{964}{15}-\dfrac{5}{4}\ln 3$	A1
		(5)
		Total 14